

Convergence in distribution is stable under small-probability modifications

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Abstract

A generic probabilistic lemma (Astra #13, route R1) needed for the division step of the posterior ratio. If for every $\eta > 0$ there are random variables $Y_n^\eta \Rightarrow W^\eta$ with $\mu\{X_n \neq Y_n^\eta\} \leq \eta$ eventually and $\mu'\{Z \neq W^\eta\} \leq \eta$, then $X_n \Rightarrow Z$ (`tendstoInDistribution_of_approx`). The workhorse is the modification bound $|\int f \circ X - \int f \circ Y| \leq C\eta$ for a test function with $\text{dist}(f(x), f(y)) \leq C$ (`abs_integral_comp_sub_le_of_ne`); the conclusion follows from Mathlib's bounded-Lipschitz characterisation of weak convergence. Zero `sorry/axiom`.

1 The things formalised

```
variable {E} [PseudoEMetricSpace E] [MeasurableSpace E] [OpensMeasurableSpace E]
[MeasurableEq E]
theorem integrable_comp_of_bounded (f : E → ℝ) (hf : Measurable f) (x : E) (C)
(hC : x y, dist (f x) (f y) ≤ C) (X : Ω → E) (hX : Measurable X) :
Integrable (fun _ => f (X _))
theorem abs_integral_comp_sub_le_of_ne (f) (hf) (x) (C) (hC) (X Y : Ω → E) (hX hY) ()
(h : 0 ≤ C) (hS : { | X _ Y _ | ENNReal.ofReal }) :
| (∫, f (X _)) - (∫, f (Y _)) | ≤ C *
theorem tendstoInDistribution_of_approx [Nonempty E] [1.IsCountablyGenerated]
(X : Ω → E) (Z : Ω' → E) (hX : n, Measurable (X n)) (hZ : Measurable Z)
(happrox : , 0 < → (Y : Ω → E) (W : Ω' → E),
(n, Measurable (Y n)) Measurable W
TendstoInDistribution Y 1 W (fun _ => ) '
(n in 1, { | X n Y n | ENNReal.ofReal })
' { | Z W | ENNReal.ofReal }) :
TendstoInDistribution X 1 Z (fun _ => ) '
```

2 Mathematics

Weak convergence of the laws is equivalent to $\int f d\mu_n \rightarrow \int f d\mu$ for every bounded Lipschitz f (Mathlib `tendsto_iff_forall_lipschitz_integral_tendsto`, which is where the countably generated filter enters). For such f with oscillation $\leq C$, changing a random variable on an event of probability $\leq \eta$ changes $\int f \circ X$ by at most $C\eta$: the integrand difference is dominated by C times the indicator of the disagreement event. Given $\varepsilon > 0$ take $\eta = \varepsilon/(3(C+1))$ and the approximants for this η ; the three terms of the triangle inequality are each $< \varepsilon/3$ eventually.

3 Paper-facing meaning

The posterior ratio $Z_n[\varphi]/Z_n[1]$ is a quotient of two jointly convergent normalised chart integrals whose limiting denominator is a.s. positive but not bounded below. Mathlib’s continuous mapping theorem needs global continuity, which division lacks; the standard fix is to divide by $\max(v, c)$ (continuous), which agrees with division except on $\{V_n < c\}$, an event of small probability uniformly in n by portmanteau. The present lemma converts that approximation into convergence in distribution of the ratio itself (next unit).

4 Lean notes

- The disagreement event $\{X \neq Y\}$ is measurable via `measurableSet_eq_fun`, which needs the class `MeasurableEq E` (instances: second-countable T_2 spaces, standard Borel spaces); we assume it on E .
- `Integrable.of_bound` with the bound $\|f(x_0)\| + C$; the indicator bound integrates via `integral_indicator_const` and `ENNReal.toReal_le_of_le_ofReal`.
- Rewriting `with tendsto_iff_forall_lipschitz_integral_tendsto` on the `TendstoInDistribution.tendsto` field needs `set_option backward.isDefEq.respectTransparency.types false` (as in Mathlib’s own proofs), because `ProbabilityMeasure` is a subtype behind a definition; then `ProbabilityMeasure.coe_mk` and `integral_map` expose the integrals over Ω .
- `field_simp` closed the scalar identity by itself (a trailing `ring` errors with “no goals”); `add_le_add (abs_add_le _ _) le_rfl` for the right-associated triangle inequality (`add_le_add_right` adds on the left).